Skyler Wu

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United States Citizen

EDUCATION

Stanford University — September 2024 - May 2029 (Expected)
Ph.D. Candidate in Statistics — GPA: 3.93
National Science Foundation Graduate Research Fellow
Advisors: Scott W. Linderman and Emmanuel J. Candès

Harvard University — September 2020 - May 2024
A.B. in Statistics & Mathematics (Summa Cum Laude with Highest Honors) — GPA: 4.00
S.M. in Applied Mathematics (Concurrent) — GPA: 3.95
Secondary Field in Computer Science
Advisor: Samuel Kou

JOURNAL AND PEER-REVIEWED CONFERENCE PUBLICATIONS

- **Skyler Wu,** Yash Nair, Emmanuel J. Candès, "Efficient Evaluation of LLM Performance with Statistical Guarantees." To appear at *NeurIPS (2026)*.
- **Skyler Wu,** Shihao Yang, Samuel Kou, "Are Statistical Methods Obsolete in the Era of Deep Learning? A Study of ODE Inverse Problems." *The American Statistician* (2026).
- David M. Zoltowski*, **Skyler Wu***, Xavier Gonzalez, Leo Kozachkov, Scott W. Linderman, "Parallelizing MCMC Across the Sequence Length." *Advances in Neural Information Processing Systems, 2025.*
- **Skyler Wu***, Austin Meyer*, Leonardo Clemente, Lucas M. Stolerman, Fred Lu, Atreyee Majumder, Rudi Verbeeck, Serge Masyn, Mauricio Santillana, "Ensemble Approaches for Short-Term Dengue Fever Forecasts: A Global Evaluation Study." *Proceedings of the National Academy of Sciences 122, 2025.*
- **Skyler Wu**, Fred Lu, Edward Raff, James Holt, "Stabilizing Linear Passive-Aggressive Online Learning with Weighted Reservoir Sampling." *Advances in Neural Information Processing Systems, 2024*.

PEER-REVIEWED WORKSHOP PAPERS

- **Skyler Wu***, Eric M. Shen*, Charumathi Badrinath*, Jiaqi Ma, Himabindu Lakkaraju, "Analyzing Chain-of-Thought Prompting in Large Language Models via Gradient-based Feature Attributions." *Challenges in Deployable Generative AI Workshop at ICML 2023.*
- **Skyler Wu,** Fred Lu, Edward Raff, James Holt, "Exploring the Sharpened Cosine Similarity." *I Can't Believe It's Not Better Workshop at NeurIPS 2022.*

WORKING PAPERS AND PREPRINTS

- **Skyler Wu**, Yash Nair, Antoine Maechler, Emmanuel J. Candès, "Pro-Active Inference." Working paper (2026).
- **Skyler Wu*,** Aymen Echarghaoui*, "Intelligently Weighting Multiple Reference Models for Direct Preference Optimization of LLMs." Working paper (2025).
- Kyla Chasalow*, **Skyler Wu*,** Susan Murphy, "Missing Data Multiple Imputation for Tabular Q-Learning in Online RL." Working paper (2025).

(* Equal Contribution)

TECHNICAL AND NON-TECHNICAL SKILLS

- **Programming:** Proficient in Python (incl. Multiprocessing, JAX, PyTorch/PyG, TensorFlow/TensorFlow Probability, HuggingFace family, NumPy, Pandas, Matplotlib/Seaborn, SciPy, scikit-learn, CVXPY, etc.), experience working in R, Bash, HTML/CSS/JavaScript, and C. Also experienced with writing and profiling custom GPU kernels in Pallas (JAX DSL) and NVIDIA's Nsight suite.
- **Foreign Languages:** Mandarin Chinese (native proficiency).

PROFESSIONAL EXPERIENCES

Booz Allen Hamilton Remote

Analytics Fellow, Junior September 2021 – October 2025

Spearheaded two ML projects for the Laboratory for Physical Sciences client. Select technical contributions include:

- Leveraged high-performance computing and scaled deep-learning architectures across multiple GPU processors for efficient running of experiments.
- Designed efficient pipelines for finding-and-replacing of target components within arbitrary neural network architectures, as well as routines for simulating adversarial attacks and investigating model interpretability.
- Stabilized passive-aggressive online learning algorithms' out-of-sample performances to near-oracle levels and with orders-of-magnitude faster wall-clock time.

Collaborated with technical teammates and client stakeholders to scope research questions, coordinate experimental work, and communicate findings.

AWARDS AND FUNDING

- **Statistics Department Teaching Assistant Award:** for excellent service and dedication to the students of Leland Stanford Junior University (June 2026).
- **National Science Foundation Graduate Research Fellowship Program:** provides 3 years of support over a 5-year fellowship period for the graduate education of individuals who have demonstrated their potential for significant research achievements in STEM or STEM education (April 2024).
- **Harvard Department of Statistics Senior Concentrator Award:** awarded annually to the graduating senior concentrator who has the best overall performance (as indicated by coursework results and thesis) and who has contributed significantly to the department (May 2024).
- **Harvard University Sophia Freund Prize:** awarded annually to the highest ranking undergraduate as determined at the final degree meeting of the Faculty. The award will be made to that student graduating summa cum laude who has the highest grade point average (May 2024).
- **Phi Beta Kappa Harvard College:** elected as a member of Harvard College's chapter of Phi Beta Kappa, Alpha Iota of Massachusetts. Elected as a member of the Class of 2024 Senior 48 (Fall 2023).
- **The Derek Bok Center for Teaching and Learning:** 2x Certificate of Distinction in Teaching (Fall 2021, Spring 2022). This certificate is awarded to course staff who earn an overall score of 4.5 out of 5.0 or higher (with a minimum of five evaluations) on the Q evaluations completed by students in their courses. Award discontinued starting Fall 2022.

TALKS AND PRESENTATIONS

Invited

- "Efficient Evaluation of LLM Performance with Statistical Guarantees." *Stanford University, Department of Statistics, Industrial Affiliates Annual Conference*. November 5, 2026.
- "An Introduction to Markov Chain Monte Carlo." *Guest lecture for STATS 217: Introduction to Stochastic Processes I.* July 8, 2026.
- "Parallelizing MCMC Across the Sequence Length." *Stanford University, Department of Statistics, Palacios Group Meeting.* May 22, 2026.
- "Ensemble Approaches for Robust and Generalizable Short-Term Forecasts of Dengue Fever." *Oklahoma State University Mathematics, Computational and Applied Math Seminar.* February 20, 2025.

Contributed

- "Parallelizing MCMC Across the Sequence Length." *Emerging Directions in Probabilistic Modeling Workshop Poster Session*, co-sponsored by the Flatiron Institute and IBM. October 5, 2026.
- "Parallelizing MCMC Across the Sequence Length: T Samples in $O(\log^2 T)$ Time." *2026 ISBA World Meeting Poster Session 2.* June 30, 2026.
- "Pro-Active Inference." *AI+Science: Accelerating Discovery Conference Poster Session 2,* co-sponsored by Stanford HAI and Stanford Data Science. May 5, 2026.
- "Pushing the Feasibility Frontier of MCMC with Parallelization Across Chain Length." *AI+Science: Accelerating Discovery Conference Poster Session 1*, co-sponsored by Stanford HAI and Stanford Data Science. May 5, 2026.
- "Efficient Evaluation of LLM Performance with Statistical Guarantees." *Stanford Causal Science Conference: Frontiers in AI Evaluation Poster Session.* April 24, 2026.

- "Parallelizing MCMC Across the Sequence Length", *Stanford-Berkeley Joint Colloquium Poster Session.* October 21, 2025.
- "Stabilizing Linear Passive-Aggressive Online Learning with Weighted Reservoir Sampling", *Stanford-Berkeley Joint Colloquium Poster Session*. October 29, 2024.

Discussant

- "Uncertainty Intervals for Ranking, with Applications in Machine Learning," by Yuval Benjamini. *International Seminar on Selective Inference.* March 19, 2026.

TEACHING EXPERIENCES

As Instructor, Stanford University

- STATS 302: Qualifying Exam Workshop (Applied Statistics) — Summer 2026

As Teaching Assistant, Stanford University

- STATS 60: Introduction to Statistics — Spring 2026
- STATS 118: Probability Theory for Statistical Inference — Fall 2025
- STATS 202: Statistical Learning and Data Science — Fall 2024

As Co-Head Teaching Fellow, Harvard University

- COMPSCI 181: Introduction to Machine Learning — Fall 2022 - Spring 2023

As Teaching Fellow, Harvard University

- STAT 111: Introduction to Statistical Inference — Spring 2024
- STAT 110: Introduction to Probability[1] — Fall 2021, 2022, 2023
- COMPSCI 181: Introduction to Machine Learning[2] — Spring 2022

SERVICE

ICML 2026 — Remote
Reviewer — February - April 2026
Reviewed 6 papers for the 2026 main conference. Designated as a Gold Reviewer (top 25% of reviewers).

Workshop for Reproducible AI @ AAAI — Remote
Reviewer — November 2025
Reviewed 3 papers for the 2026 workshop.

I Can't Believe It's Not Better Workshop @ NeurIPS 2023 — Remote
Reviewer — October 2023
Reviewed 3 papers for the 2023 workshop iteration, focusing on "Failure Modes in the Age of Foundation Models."

[1] See sample review lectures and new Teaching Fellow on-boarding curriculum that I wrote from scratch.
[2] See custom Python/NumPy and scikit-learn/PyTorch guides that I created from scratch.